

■ Bems Farms — Driver Mobile App API Specification

Official Engineering Guide for Mobile App Developers (React Native, Flutter, iOS & Android)

Target Mobile Platforms: React Native / Flutter / iOS / Android	Authentication: JWT Bearer (`Authorization: Bearer <token>`)
Production API URL: https://api.bemsfarms.com/api/driver	Local Development URL: http://localhost:5000/api/driver
Payload Standard: application/json	Driver Wallet & DVA: Monnify Virtual NUBAN + Zone-Based Earnings

1. Driver API Directory & Endpoint Index

Category	Method & Endpoint	Auth	Description & Mobile Trigger
Onboarding	GET /onboarding/verify?token=...	Public	Verify invite token & pre-populate driver signup details
Onboarding	POST /onboarding/submit	Public	Submit driver KYC, NIN, vehicle details & settlement bank
Auth	POST /auth/login	Public	Driver email/phone + password authentication & JWT token
Auth	POST /auth/forgot-password	Public	Send password reset token via SMS / Email
Auth	POST /auth/reset-password	Public	Confirm reset token and configure new driver password
Profile	GET /auth/me	Bearer	Retrieve driver profile, rating, stats & Monnify Virtual Account
Profile	PATCH /auth/profile	Bearer	Update driver phone number, profile photo, or bank details
Availability	PATCH /availability	Bearer	Toggle duty state: Online (ready for dispatch) vs Offline
Deliveries	GET /deliveries	Bearer	Fetch assigned active deliveries with customer addresses & items
Deliveries	GET /deliveries/:orderId	Bearer	Fetch full itemized breakdown for a specific delivery drop
Milestones	PATCH /deliveries/:orderId/status	Bearer	Progress milestone (accepted, en_route, arrived, delivered)
History	GET /deliveries/history	Bearer	Fetch paginated past delivery drops and completed trips
Telemetry	POST /location	Bearer	Background GPS coordinate stream ping (every 10–15s)
Wallet	GET /earnings	Bearer	Live wallet balance, Monnify DVA NUBAN, zone rate card & history
Wallet / Bank	GET /banks	Bearer	Directory of Nigerian commercial banks & fintechs with codes
Wallet / Bank	POST /bank/resolve	Bearer	Verify 10-digit NUBAN account number & resolve legal account name
Wallet	POST /withdraw	Bearer	Request payout of unwithdrawn earnings to personal bank account

2. Driver Authentication, Onboarding & Profile APIs

2.1 Driver Login (`POST /api/driver/auth/login`)

Authenticates driver by phone number or email and password. Returns JWT token and driver profile with Dedicated Virtual Account (DVA).

Request Payload	Success Response (200 OK)
<pre>{ "emailOrPhone": "08123456789", "password": "Password123!"}</pre>	<pre>{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "message": "Login successful", "driver": { "id": 6, "name": "Victor Kalu", "phone": "08123456789", "email": "victor.kalu@bemsfarms.com", "wallet_account_number": "3397202109", "wallet_bank_name": "Monnify / Wema Bank", "wallet_account_name": "BEMS - VICTOR KALU", "bank_name": "GTBank", "account_number": "0123456789", "wallet_balance": 16500.00, "is_available": true }}</pre>

2.2 Toggle Availability State (`PATCH /api/driver/availability`)

Request Payload	Success Response (200 OK)
<pre>{ "is_available": true, "latitude": 9.0765, "longitude": 7.3986 }</pre>	<pre>{ "message": "Duty status updated to Online", "is_available": true, "status": "active" }</pre>

3. Dispatch Deliveries & Milestone Progression

Drivers receive live delivery drops and progress them through 5 status milestones.

Milestone	Action Description	Customer Tracking State	Financial Trigger
accepted	Driver accepts delivery assignment	driver_assigned	—
awaiting_pickup	Driver at hub/store loading goods	packed_ready	—
en_route	Driver departed with order package	out_for_delivery	—
arrived	Driver arrived at customer destination	driver_arrived	—
delivered	Handover complete with photo proof	delivered	Auto-Credit Zone Commission
failed	Delivery attempted but customer unavailable	delivery_attempted	—

3.1 Update Milestone (`PATCH /api/driver/deliveries/:orderId/status`)

Request Payload	Success Response (200 OK)
<pre>{ "status": "delivered", "proof_note": "Handed directly to customer Mrs. Ngozi.", "proof_photo": "https://storage.googleapis.com/bems-uploads/proof_del_998.jpg" }</pre>	<pre>{ "message": "Delivery completed successfully", "delivery_status": "delivered", "order_status": "delivered", "commission_earned": 1750.00, "new_wallet_balance": 18250.00 }</pre>

4. Driver Wallet, Bank Verification & Withdrawal System

The driver wallet architecture provides real-time balances, Dedicated Virtual Accounts (Monnify DVA NUBAN), Nigerian bank directory lookup, instant NIP account name verification, and on-demand withdrawal requests.

4.1 Get Driver Wallet & Earnings (`GET /api/driver/earnings`)

Returns live wallet balance, unwithdrawn earnings, pending payout amounts, Monnify Dedicated Virtual Account (DVA), saved withdrawal bank, zone rate card, and recent transaction history.

```
{
  "wallet": {
    "total_earned": 28500.00,
    "total_paid": 12000.00,
    "pending_payouts": 0.00,
    "available_balance": 16500.00,
    "commission_per_delivery": 1750.00,
    "dedicated_virtual_account": {
      "account_number": "8559127267",
      "bank_name": "Monnify / Wema Bank",
      "account_name": "BEMS - VICTOR KALU"
    },
    "withdrawal_bank": {
      "bank_name": "GTBank",
      "account_number": "0123456789",
      "account_name": "Victor Kalu"
    }
  },
  "zone_rates": [
    { "zone_id": "ZONE001", "zone_name": "Zone 1 - Maitama & Asokoro", "driver_earning_fee": 700.00, "commission_percent": 70, "estimated_eta": "20 - 35 mins" },
    { "zone_id": "ZONE002", "zone_name": "Zone 2 - Garki, Wuse, Central Area", "driver_earning_fee": 1750.00, "commission_percent": 70, "estimated_eta": "25 - 45 mins" },
    { "zone_id": "ZONE003", "zone_name": "Zone 3 - Utako, Jabi, Mabushi", "driver_earning_fee": 2450.00, "commission_percent": 70, "estimated_eta": "30 - 50 mins" },
    { "zone_id": "ZONE004", "zone_name": "Zone 4 - Gwarinpa, Kubwa", "driver_earning_fee": 3500.00, "commission_percent": 70, "estimated_eta": "40 - 65 mins" }
  ]
}
```

```
{
  "zone_id": "ZONE005", "zone_name": "Zone 5 - Lugbe, Airport Road", "driver_earning_fee": 5600.00, "commission_percent": 70, "estimated_eta": "45 - 75 mins" },
  { "zone_id": "ZONE006", "zone_name": "Zone 6 - Interstate Express", "driver_earning_fee": 24500.00, "commission_percent": 70, "estimated_eta": "Same Day" }
}
```

4.2 List Nigerian Commercial Banks & Fintechs (`GET /api/driver/banks`)

Provides the mobile app with the complete directory of Nigerian banks and fintech institutions (Access, GTBank, Zenith, OPay, PalmPay, Kuda, Moniepoint, Wema, etc.) for populating bank picker modals.

```
{
  "status": "success",
  "count": 33,
  "banks": [
    { "name": "Access Bank", "code": "044", "ussd": "*901#" },
    { "name": "Guaranty Trust Bank (GTBank)", "code": "058", "ussd": "*737#" },
    { "name": "Kuda Bank", "code": "50211", "ussd": "" },
    { "name": "Moniepoint MFB", "code": "50515", "ussd": "" },
    { "name": "OPay Digital Services", "code": "999992", "ussd": "" },
    { "name": "PalmPay", "code": "999991", "ussd": "" },
    { "name": "United Bank for Africa (UBA)", "code": "033", "ussd": "*919#" },
    { "name": "Wema Bank / ALAT", "code": "035", "ussd": "*945#" },
    { "name": "Zenith Bank", "code": "057", "ussd": "*966#" }
  ]
}
```

4.3 Resolve & Verify Bank Account Name (`POST /api/driver/bank/resolve`)

Validates a 10-digit NUBAN account number against the selected bank code using NIP name enquiry. The mobile app calls this as soon as the driver enters 10 digits to display the verified account holder name.

Request Payload	Success Response (200 OK)
<pre>{ "account_number": "0123456789", "bank_code": "058", "bank_name": "Guaranty Trust Bank (GTBank)" }</pre>	<pre>{ "status": "success", "account_number": "0123456789", "bank_code": "058", "bank_name": "Guaranty Trust Bank (GTBank)", "account_name": "VICTOR KALU", "is_valid": true, "message": "Account name successfully resolved: VICTOR KALU" }</pre>

4.4 Request Withdrawal (`POST /api/driver/withdraw`)

Submits a withdrawal request of earnings directly to the driver's verified commercial bank account. Automatically updates the driver's default saved settlement account.

Request Payload	Success Response (201 Created)
<pre>{ "amount": 10000.00, "bank_name": "Guaranty Trust Bank (GTBank)", "bank_code": "058", "account_number": "0123456789", "account_name": "VICTOR KALU", "notes": "Weekly earnings withdrawal" }</pre>	<pre>{ "message": "Withdrawal request submitted successfully.", "payout": { "payout_ref": "PAY-MNFI-99201", "amount": 10000.00, "bank_name": "Guaranty Trust Bank (GTBank)", "account_number": "0123456789", "account_name": "VICTOR KALU", "status": "pending" }, "remaining_available_balance": 6500.00 }</pre>

5. Zone-Based Compensation Rate Schedule (Zones 1–6)

Zone ID	Zone Coverage Area	Customer Delivery Fee	Driver Drop Payout (70%)	Estimated ETA
ZONE001	Zone 1 - Maitama & Asokoro	■1,000.00	■700.00	20 - 35 mins
ZONE002	Zone 2 - Garki, Wuse, Central Area	■2,500.00	■1,750.00	25 - 45 mins
ZONE003	Zone 3 - Utako, Jabi, Mabushi	■3,500.00	■2,450.00	30 - 50 mins
ZONE004	Zone 4 - Gwarinpa, Kubwa Expressway	■5,000.00	■3,500.00	40 - 65 mins
ZONE005	Zone 5 - Lugbe, Airport Road	■8,000.00	■5,600.00	45 - 75 mins
ZONE006	Zone 6 - Interstate / Regional Express	■35,000.00	■24,500.00	Same Day / Next Day

6. Mobile Client Engineering Checklist

- **1. Secure Storage:** Store the JWT token securely using SecureStore (Expo) / EncryptedSharedPreferences (Android) / Keychain (iOS).
- **2. Background GPS Streaming:** When online (is_available: true), stream background coordinates to POST /api/driver/location every 10–15s.
- **3. Navigation Deep Linking:** Deep-link customer coordinates to Google Maps (geo:\${lat},\${lng}?q=\${lat},\${lng}) or Apple Maps (maps://?daddr=\${lat},\${lng}).
- **4. Direct Customer Contact:** Provide 1-tap dialer (tel:\${phone}) and WhatsApp quick chat (https://wa.me/234\${phone}).
- **5. Live Earnings View:** Show driver's real-time unwithdrawn earnings and Dedicated Virtual Account (DVA) directly on the app's wallet tab.
- **6. Offline Caching:** Cache active deliveries locally so drivers can view delivery addresses and items even during cellular network dips.